THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 1: **Coordinates** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 20	Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$.
2	**(c)** $(4.5,\ 4.5)$	Average the coordinates.
3	**(a)** $(-1, -3)$	The centre is the midpoint of a diameter.
4	**(c)** $(3, 0)$	Set $PA = PB$ with $P = (x, 0)$.
5	**(a)** Quadrant III	Use the sign conditions on the coordinates.
6	**(d)** 36	Add the three side lengths.
7	**(d)** $(-4, -9)$	Both signs change.
8	**(c)** 15	Area = length × breadth.
9	**(b)** 65	Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$.
10	**(c)** $(-1.5,\ 6.5)$	Average the coordinates.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$d^2 = (4 - (1))^2 + (5 - (1))^2 = 25$ (1); $d = 5$ units (1).	1+1
12	$\left(\frac{3+(6)}{2}, \frac{-4+(7)}{2}\right)$ (1); Midpoint $= (4.5,\ 1.5)$ (1).	1+1
13	Centre is the midpoint of the diameter (1); $B = (-3, -6)$ (1).	1+1
14	$d^2 = (2 - (-2))^2 + (6 - (3))^2 = 25$ (1); $d = 5$ units (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$AB = 1 \times 5 = 5$ (direction $(3, 4)$); $BC = 2 \times 5 = 10$ (1); $AC = 15 = AB + BC$ (1); So the points are collinear (1).	1+1+1
16	$(2 - k)^2 + 16 = 25$ (1); $(2 - k)^2 = 9$ (1); $k = -1$ or $k = 5$ (1).	1+1+1
17	$AB^2 = 18$, $BC^2 = 18$; $CA^2 = 36$ (1); Two sides equal (1); $AB^2 + BC^2 = CA^2$: right angle at B (1).	1+1+1
	OR: $AB = 2 \times 13 = 26$ (direction $(5, 12)$); $BC = 1 \times 13 = 13$ (1); $AC = 39 = AB + BC$ (1); So the points are collinear (1).	1+1+1
18	$(-3 - k)^2 + 144 = 169$ (1); $(-3 - k)^2 = 25$ (1); $k = -8$ or $k = 2$ (1).	1+1+1
	OR: $AB^2 = 18$, $BC^2 = 18$; $CA^2 = 36$ (1); Two sides equal (1); $AB^2 + BC^2 = CA^2$: right angle at B (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$AB^2 = 9 + 16 = 25$ (1); $BC^2 = DA^2 = CD^2 = 25$: all sides equal 5 (1); $AC = 2 \times 3 = 6$ (1); $BD = 2 \times 4 = 8$ (1); $6 \neq 8$, so not a square (1).	1+1+1+1+1
	OR: $D = A + C - B = (2, 5)$ (1); Midpoint of $AC = (3.5, 3)$ (1); Midpoint of $BD = (3.5, 3)$ (same point) (1); Diagonals bisect each other (1); Hence $ABCD$ is a parallelogram (1).	1+1+1+1+1
20	$AB^2 = 25 + 144 = 169$ (1); $BC^2 = DA^2 = CD^2 = 169$: all sides equal 13 (1); $AC = 2 \times 5 = 10$ (1); $BD = 2 \times 12 = 24$ (1); $10 \neq 24$, so not a square (1).	1+1+1+1+1
	OR: $D = A + C - B = (3, 7)$ (1); Midpoint of $AC = (4.5, 5)$ (1); Midpoint of $BD = (4.5, 5)$ (same point) (1); Diagonals bisect each other (1); Hence $ABCD$ is a parallelogram (1).	1+1+1+1+1

Mistakes to avoid:

- Writing the point as (y, x): the abscissa x always comes first, so $(2, 3) \neq (3, 2)$.
- Mixing up the signs of the quadrants, e.g. placing $(-3, 4)$ in Quadrant IV; Quadrant II is $(-, +)$.
- Reflecting in the x-axis but changing the sign of the abscissa (that is a reflection in the y-axis).
- Saying that $(0, -6)$ or $(5, 0)$ lies in a quadrant; points on the axes belong to no quadrant.
- In the distance formula, forgetting the square root or adding the coordinates instead of subtracting them.
- In the midpoint formula, dividing only one coordinate by 2, or subtracting instead of adding.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 1: **Coordinates** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** $(3, 0)$	The centre is the midpoint of a diameter.
2	**(a)** $(-2, 0)$	Set $PA = PB$ with $P = (x, 0)$.
3	**(a)** Quadrant II	Use the sign conditions on the coordinates.
4	**(c)** 70	Add the three side lengths.
5	**(c)** $(6, -2)$	Both signs change.
6	**(b)** 35	Area = length × breadth.
7	**(c)** 137	Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$.
8	**(c)** $(2.5,\ 2.5)$	Average the coordinates.
9	**(b)** $(0, 5)$	The centre is the midpoint of a diameter.
10	**(d)** $(4, 0)$	Set $PA = PB$ with $P = (x, 0)$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\left(\frac{-5+(2)}{2}, \frac{2+(5)}{2}\right)$ (1); Midpoint $= (-1.5,\ 3.5)$ (1).	1+1
12	Centre is the midpoint of the diameter (1); $B = (5, 10)$ (1).	1+1
13	$d^2 = (-3 - (0))^2 + (1 - (5))^2 = 25$ (1); $d = 5$ units (1).	1+1
14	$\left(\frac{7+(-4)}{2}, \frac{1+(-8)}{2}\right)$ (1); Midpoint $= (1.5,\ -3.5)$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$AB = 1 \times 17 = 17$ (direction $(8, 15)$); $BC = 3 \times 17 = 51$ (1); $AC = 68 = AB + BC$ (1); So the points are collinear (1).	1+1+1
16	$(5 - k)^2 + 225 = 289$ (1); $(5 - k)^2 = 64$ (1); $k = -3$ or $k = 13$ (1).	1+1+1
17	$AB^2 = 18$, $BC^2 = 18$; $CA^2 = 36$ (1); Two sides equal (1); $AB^2 + BC^2 = CA^2$: right angle at B (1).	1+1+1
	OR: $AB = 3 \times 5 = 15$ (direction $(3, 4)$); $BC = 1 \times 5 = 5$ (1); $AC = 20 = AB + BC$ (1); So the points are collinear (1).	1+1+1
18	$(0 - k)^2 + 576 = 625$ (1); $(0 - k)^2 = 49$ (1); $k = -7$ or $k = 7$ (1).	1+1+1
	OR: $AB^2 = 8$, $BC^2 = 8$; $CA^2 = 16$ (1); Two sides equal (1); $AB^2 + BC^2 = CA^2$: right angle at B (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$AB^2 = 64 + 225 = 289$ (1); $BC^2 = DA^2 = CD^2 = 289$: all sides equal 17 (1); $AC = 2 \times 8 = 16$ (1); $BD = 2 \times 15 = 30$ (1); $16 \neq 30$, so not a square (1).	1+1+1+1+1
	OR: $D = A + C - B = (0, 6)$ (1); Midpoint of $AC = (1.5, 3)$ (1); Midpoint of $BD = (1.5, 3)$ (same point) (1); Diagonals bisect each other (1); Hence $ABCD$ is a parallelogram (1).	1+1+1+1+1
20	$AB^2 = 49 + 576 = 625$ (1); $BC^2 = DA^2 = CD^2 = 625$: all sides equal 25 (1); $AC = 2 \times 7 = 14$ (1); $BD = 2 \times 24 = 48$ (1); $14 \neq 48$, so not a square (1).	1+1+1+1+1
	OR: $D = A + C - B = (5, 3)$ (1); Midpoint of $AC = (6, 1.5)$ (1); Midpoint of $BD = (6, 1.5)$ (same point) (1); Diagonals bisect each other (1); Hence $ABCD$ is a parallelogram (1).	1+1+1+1+1

Mistakes to avoid:

- Writing the point as (y, x): the abscissa x always comes first, so $(2, 3) \neq (3, 2)$.
- Mixing up the signs of the quadrants, e.g. placing $(-3, 4)$ in Quadrant IV; Quadrant II is $(-, +)$.
- Reflecting in the x-axis but changing the sign of the abscissa (that is a reflection in the y-axis).
- Saying that $(0, -6)$ or $(5, 0)$ lies in a quadrant; points on the axes belong to no quadrant.
- In the distance formula, forgetting the square root or adding the coordinates instead of subtracting them.

- In the midpoint formula, dividing only one coordinate by 2, or subtracting instead of adding.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 1: **Coordinates** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** Quadrant IV	Use the sign conditions on the coordinates.
2	**(d)** 48	Add the three side lengths.
3	**(a)** $(-8, 5)$	Both signs change.
4	**(a)** 63	Area = length × breadth.
5	**(b)** 145	Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$.
6	**(b)** $(1.5,\ 1.5)$	Average the coordinates.
7	**(d)** $(2, 12)$	The centre is the midpoint of a diameter.
8	**(b)** $(-5, 0)$	Set $PA = PB$ with $P = (x, 0)$.
9	**(c)** Quadrant I	Use the sign conditions on the coordinates.
10	**(b)** 60	Add the three side lengths.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Centre is the midpoint of the diameter (1); $B = (10, 2)$ (1).	1+1
12	$d^2 = (1 - (6))^2 + (10 - (-2))^2 = 169$ (1); $d = 13$ units (1).	1+1
13	$\left(\frac{1+(4)}{2}, \frac{6+(-3)}{2}\right)$ (1); Midpoint $= (2.5,\ 1.5)$ (1).	1+1
14	Centre is the midpoint of the diameter (1); $B = (-8, -4)$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$AB = 1 \times 13 = 13$ (direction $(5, 12)$); $BC = 3 \times 13 = 39$ (1); $AC = 52 = AB + BC$ (1); So the points are collinear (1).	1+1+1
16	$(4 - k)^2 + 64 = 100$ (1); $(4 - k)^2 = 36$ (1); $k = -2$ or $k = 10$ (1).	1+1+1
17	$AB^2 = 18$, $BC^2 = 18$; $CA^2 = 36$ (1); Two sides equal (1); $AB^2 + BC^2 = CA^2$: right angle at B (1).	1+1+1
	OR: $AB = 2 \times 25 = 50$ (direction $(7, 24)$); $BC = 1 \times 25 = 25$ (1); $AC = 75 = AB + BC$ (1); So the points are collinear (1).	1+1+1
18	$(-1 - k)^2 + 144 = 225$ (1); $(-1 - k)^2 = 81$ (1); $k = -10$ or $k = 8$ (1).	1+1+1
	OR: $AB^2 = 18$, $BC^2 = 18$; $CA^2 = 36$ (1); Two sides equal (1); $AB^2 + BC^2 = CA^2$: right angle at B (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$AB^2 = 400 + 441 = 841$ (1); $BC^2 = DA^2 = CD^2 = 841$: all sides equal 29 (1); $AC = 2 \times 20 = 40$ (1); $BD = 2 \times 21 = 42$ (1); $40 \neq 42$, so not a square (1).	1+1+1+1+1
	OR: $D = A + C - B = (0, 3)$ (1); Midpoint of $AC = (0.5, 0)$ (1); Midpoint of $BD = (0.5, 0)$ (same point) (1); Diagonals bisect each other (1); Hence $ABCD$ is a parallelogram (1).	1+1+1+1+1
20	$AB^2 = 36 + 64 = 100$ (1); $BC^2 = DA^2 = CD^2 = 100$: all sides equal 10 (1); $AC = 2 \times 6 = 12$ (1); $BD = 2 \times 8 = 16$ (1); $12 \neq 16$, so not a square (1).	1+1+1+1+1
	OR: $D = A + C - B = (3, 8)$ (1); Midpoint of $AC = (3.5, 5)$ (1); Midpoint of $BD = (3.5, 5)$ (same point) (1); Diagonals bisect each other (1); Hence $ABCD$ is a parallelogram (1).	1+1+1+1+1

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